

Laboratory Manual

Electrical Circuits I Laboratory

EEE/ECE 202 (V1, V2)

EEE/ECE 101L (V3)

Department of Electrical & Electronic Engineering (EEE)

School of Engineering (SoE)

Brac University



Revision: July 2022

A. Course Objectives:

The objectives of this course are to:

- train students to construct, test and debug DC circuits;
- train students to simulate DC circuits using PSpice.

B. Course Outcomes, CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
EEE_ECE 101 / EEE_ECE 201 Electrical Circuits I					
CO1	Explain the fundamental concepts of linear electrical circuit elements and magnetic properties of materials	a	Cognitive/ Understand	Lectures, notes	Assignment, Quiz
CO2	Apply different circuit analysis techniques and circuit theorems to solve circuits for unknown quantities	a	Cognitive/ Apply	Lectures, notes	Assignment, Quiz, Exam
CO3	Interpret the natural and transient responses of 1st order electric systems involving capacitors and inductors	a	Cognitive/ Apply	Lectures, notes	Assignment, Quiz, Exam
EEE_ECE101L / EEE_ECE 202 Electrical Circuits I Lab					
CO4	Use simulation tool to construct DC circuit in schematic level	e	Cognitive/ Understand, Psychomotor/ Manipulation	Lab Class	Project
CO5	Demonstrate basic proficiency in building, debugging and testing basic electrical circuits	e	Psychomotor/ Manipulation	Lab Class	Project

C. Mark Distribution

Assessment Tools	Weightage
Attendance	10%
Lab Work (Hardware)	10%
Lab Report (Hardware, Software)	20%
Project	20%
Lab Exam (Hardware)	20%
Lab Exam (Software)	20%

D. References

Sl.	Title	Author(s)	Publication	Edition	Publisher	ISBN
1	Engineering Circuit Analysis	W. H. Hayt, J. Kemmerly and S. M. Durbin	2007	8 th Ed.	McGraw-Hill	978-0-07-352957-8
2	Introductory Circuit Analysis	Robert L. Boylestad	2015	11 th Ed.	Prentice-Hall	0-13-173044-4

Lab Safety and Security Issues

1. Laboratory Safety Rules (General Guidelines):

The Department of EEE maintains general safety rules for laboratories. The guideline is attached in front of the door in each of the laboratories. The written rules are as follows.

- a. Closed shoes must be worn that will provide full coverage of the feet and appropriate personnel clothing must be worn.
- b. Always check if the power switch is off before plugging in to the outlet. Also, turn the instrument or equipment OFF before unplugging from the outlet.
- c. Before supplying power to the circuit, the connections and layouts must be checked by the teacher.
- d. Voltage equal or above 50V are always dangerous. Therefore, extra precautions must be taken as voltage level is increased.
- e. Extension cords should be used only when necessary and only on a temporary basis.
- f. Once the lab exercise is done, all equipment must be powered down and all probes, cords and other instruments must be returned to their proper position.
- g. In case of fire, disconnect the electrical mains power source if possible.
- h. Students must be familiar with the locations and operations of safety and emergency equipment like Emergency power off, Fire alarm switch and so on.
- i. Eating, drinking, chewing gum inside electrical laboratories are strictly prohibited.
- j. Do not use damaged cords or cords that become too hot or cords with exposed wiring and if something like that is found, inform the teacher/LTO right away.
- k. No laboratory equipment can be removed from their fixed places without the teacher/LTO's authorization.
- l. No lab work must be performed without the laboratory teacher/lab technical officer being present.

2. Electrical Safety

To prevent electrical hazards, there are symbols in front of the Electrical Distribution Board, High voltage three phase lines in the lab, Backup generator and substation. Symbols related to Arc Flash and Shock Hazard, Danger: High Voltage, Authorized personnel Only, no smoking etc. are posted in required places. Only authorized personnel are allowed to open the distribution boxes.

3. Electrical Fire:

If an electrical fire occurs, try to disconnect the electrical power source, if possible. If the fire is small, you are not in immediate danger, use any type of fire extinguisher except water to extinguish the fire. When in doubt, push in the Emergency Power Off button.

4. IMPORTANT:

Do not use water on an electrical fire.

List of Activities

Week	Activity	Page	Is this activity used for any CO assessment?	
			Yes	No
1	Group Formation, Marks Distribution, Report Structure, Project Requirement, Activity List, Course Outcome Assessment, Student Feedback, Google Classroom etc.	i		
2	Experiment 1 – Verification of KVL and KCL	1		
3	Experiment 2 – Introduction to Series-Parallel Circuits	9		
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6	Experiment 5 – Verification of Thevenin's Theorem and Maximum Power Transfer Theorem	17		
7	Recess for Midterm Examination of Theory Courses			
8	Experiment 6 – Generation and Measurement of Waveforms	24		
9	Experiment 7 – Study of Transient Behavior of RC Circuit	30		
10	Experiment 8 – Verification of Thevenin's Theorem and Maximum Power Transfer Theorem and Study of Transient Behavior of RC Circuit using PSpice	23		
11	Hardware Examination			
12	Project Submission	36	✓	

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Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic Engineering (EEE)

Brac University

Experiment No. 1

Verification of KVL and KCL

Part-A: Verification of KVL

1. **Objective:** This experiment is intended to verify Kirchhoff's voltage law (KVL) with the help of series circuits.
2. **Theoretical Background:** KVL states that around any closed circuit the algebraic sum of the voltage rises equals the algebraic sum of the voltage drops.
3. **Equipment:**
 - One multimeter
 - Three Resistors
 - One DC power supply

4. Circuit Diagram:

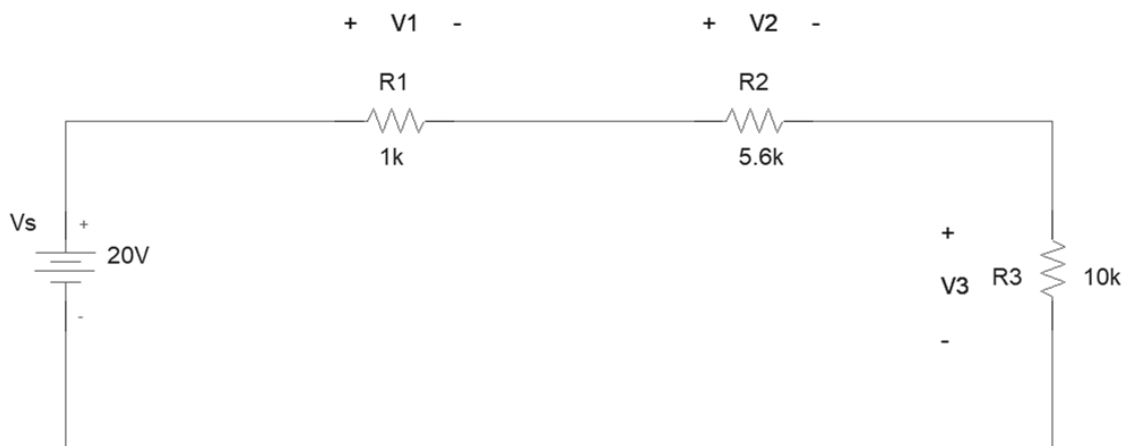


Fig. 1: Series circuit to demonstrate KVL

5. Procedure:

- Connect the resistors R_1 , R_2 and R_3 in series to a DC power supply as shown in Fig. 1.
- Take readings of V_1 , V_2 , V_3 , V_s using a multimeter.
- Verify KVL as $V_s = V_1 + V_2 + V_3$.

6. Data Table:

Verification of KVL:

V_s (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1 + V_2 + V_3$ (V)

Faculty Signature and Date

7. Lab report directions:

- State the rules of connecting voltmeter and ammeter in the circuit.
- How do you calculate the power dissipation of a resistor?

Part-B: Verification of KCL

1. Objective: This experiment is intended to verify Kirchhoff's current law (KCL) with the help of a series-parallel circuit.

2. Theoretical Background: KCL states that the algebraic sum of the currents entering any node equals the sum of the currents leaving the node.

3. Equipment:

- Three resistors
- One multimeter
- One DC supply

4. Circuit Diagram:

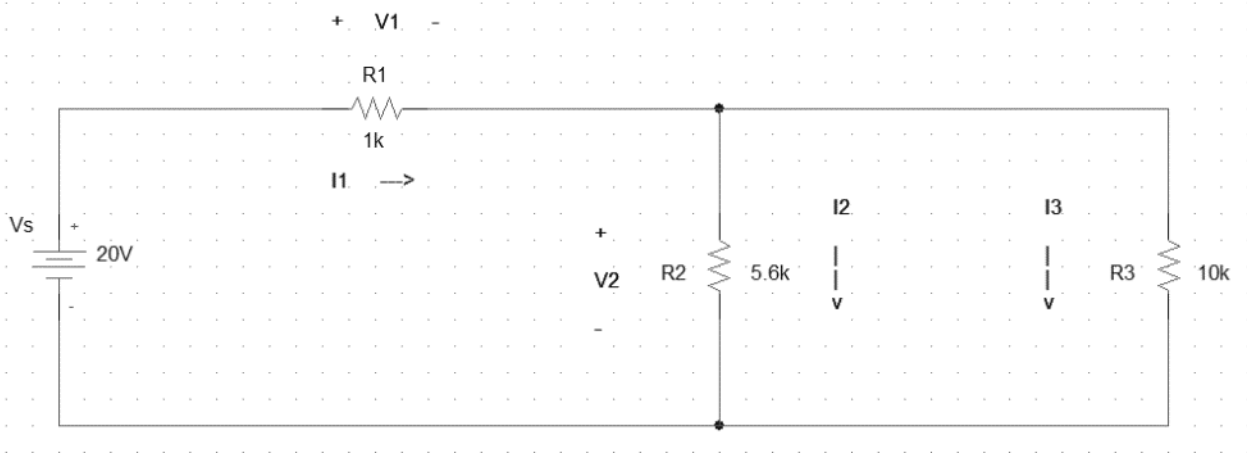


Fig. 2: Verification of KCL

5. Procedure:

- i. Connect the resistors in series-parallel across the power supply as shown in Fig. 2.
- ii. Measure the voltage drop across the resistors.
- iii. Calculate the currents I_1 , I_2 and I_3 by using $I_1 = V_1/R_1$, $I_2 = V_2/R_2$ and $I_3 = V_2/R_3$ as given above and prove that $I_1 = I_2 + I_3$.

6. Data Table:

Verification of KCL:

R_1 (k Ω)	R_2 (k Ω)	R_3 (k Ω)	V_1 (V)	V_2 (V)	$I_1=V_1/R_1$ (mA)	$I_2=V_2/R_2$ (mA)	$I_3=V_2/R_3$ (mA)	I_2+I_3 (mA)

Faculty Signature and Date

7. Lab report directions:

1. Calculate the total power supplied by the source and power absorbed by each of the resistor.
2. Is the supplied power equal to total power absorbed?

Software Simulation (Experiment No. 1 Cont'd)

1. Objective: This part is intended to illustrate the simulation procedures for determining voltages and currents by using PSpice software.

2. Problem statement: Use PSpice software for determining voltages and currents.

3. Software and Device requirement:

PSpice

Minimum PC specifications:

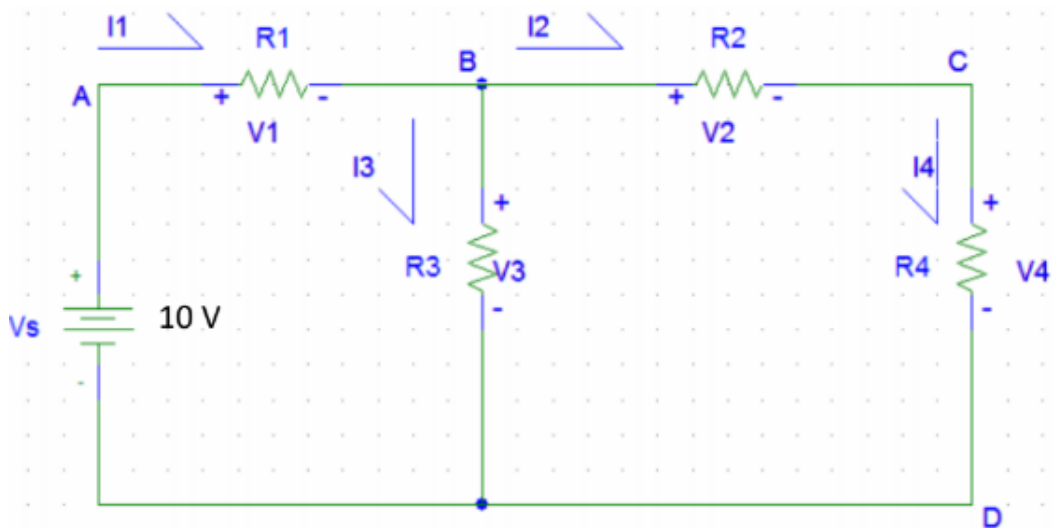
Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit); Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

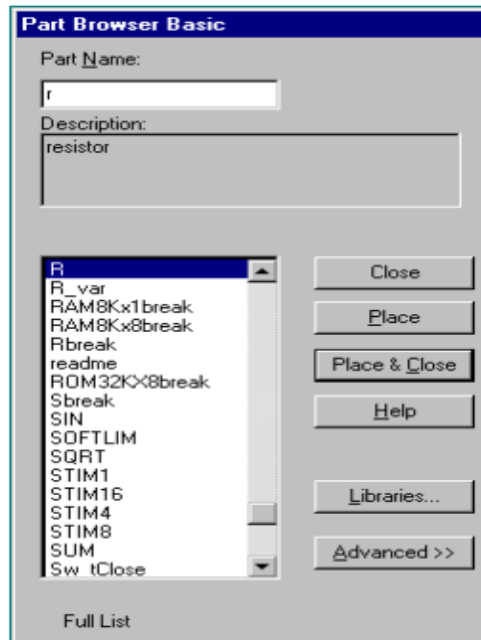
Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

4. Circuit Diagram:



5. Procedure:

i) Open PSpice schematics. The first thing that you have to do is get some or all of the parts you need to simulate your circuit. This can be done by Clicking on the 'get new parts' button , or Pressing "Control+G", or
Going to "Draw" and selecting "Get New Part..."



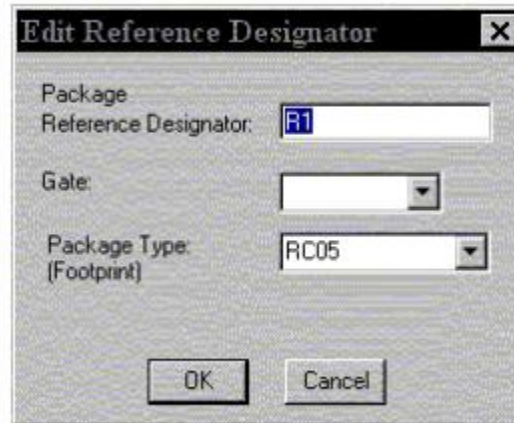
ii) Once this box is open, select a part that you want in your circuit. This can be done by typing in the name or scrolling down the list until you find it. Some common parts are: r – resistor, GND_ANALOG or GND_EARTH—this is very important, you MUST have a ground in your circuit, VAC and VDC – voltage sources, IAC and DC – current sources.

iii) Upon selecting your parts, click on the place button then click where you want it to be placed. Once you have all the parts you think you need, close that box. You can always open it again if you need more or different parts.

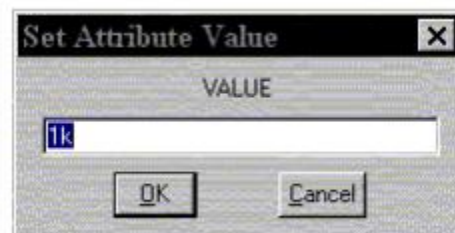
iv) Now put them in the places that make the most sense (usually a rectangle works well for simple circuits). Just select the part and drag it where you want it. To rotate parts so that they will fit in your circuit nicely, click on the part and press “Ctrl+R” (or edit “Rotate”). To flip them, press “Ctrl+F” (or Edit “Flip”). If you have any parts left over, just select them and press “Delete”.

v) Now you’ll have to connect the parts with wires. Go up to the tool bar and select “Draw Wire” or “Ctrl+W” or go to “Draw” and select “Wire”. With the pencil looking pointer, click on one end of a part, when you move your mouse around, you should see dotted lines appear. Attach the other end of your wire to the next part in the circuit. Repeat this until your circuit is completely wired. To get rid of the pencil, right click or press “Esc”. If the wire doesn’t go the way you want (it doesn’t look the way you want), you can make extra bends in it by clicking in different places on the way (each click will form a corner).

vi) Changing the Name of the Part: To change the name, double click on the present name (C1, or R1 or whatever your part is), then a box will pop up (Edit Reference Designator). In the top window, you can type in the name you want the part to have. Please note that if you double click on the part or its value, a different box will appear.

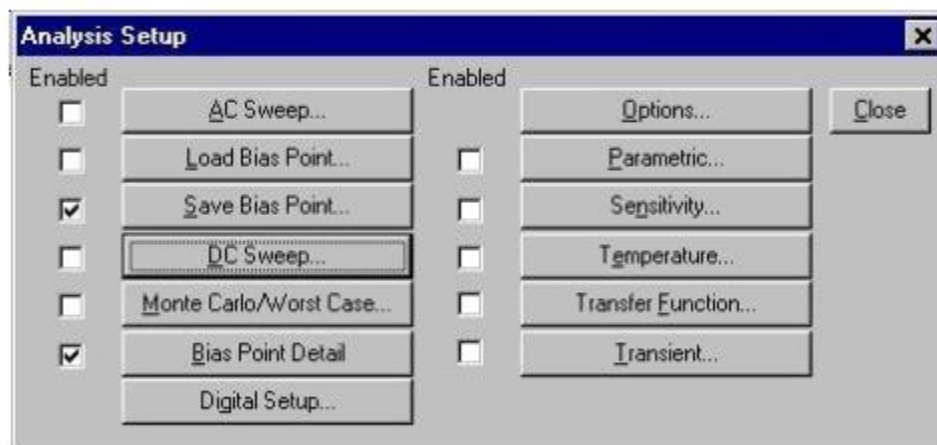


vii) Changing the Value of the Part: If you only want to change the value of the part, you can double click on the present value and a box called “Set Attribute Value” will appear. Type in the new value and press OK.

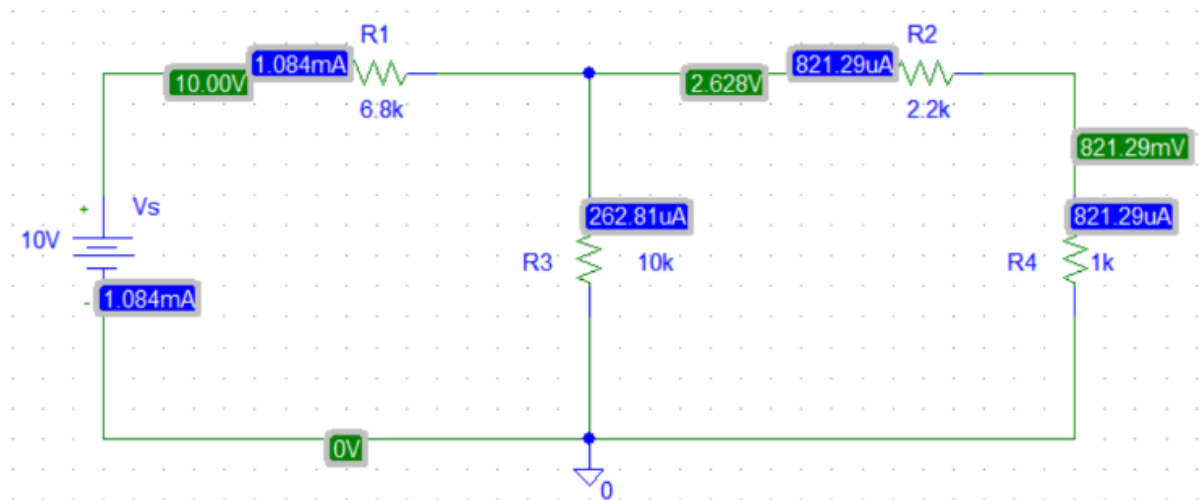


viii) Voltage and Current Bubbles setting: These are important if you want to measure the voltage at a point or the current going through that point. To add voltage or current bubbles, go to the right side of the top tool bar and select “Voltage/Level Marker” (Ctrl+M) or “Current Marker” . To get either of these, go to “Markers” and either “Voltage/Level Marker” or “Current Marker”.

ix) Open the analysis menu. Enable the appropriate analysis options and the press close.



x) Click on the Simulate button on the tool bar (or Analysis, Simulate, or F11). The results should be available now.



6. Data Table:

V_1 (V)	V_2 (V)

I_1 (mA)	I_2 (mA)	I_3 (mA)

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Experiment No. 2

Introduction to Series-Parallel Circuits

1. **Objective:** The experiment is to acquaint the students with series-parallel circuits and to give them the idea about how to connect different circuits in bread board.
2. **Theoretical Background:** An electrical circuit is a continuous path or array of paths through which an electrical current can flow. The two different ways in which components of a circuit can be connected are called "series" and "parallel". In a series connection, components are connected one after another; therefore, the same current flows through all of them. In a parallel connection, the circuit components are connected side by side. That is, the positive and negative sides of each component are respectively connected together; therefore, each has the same potential drop across. In this lab, we will explore measurements of current and potential difference in simple circuits.
3. **Equipment:**
 - DC power supplies
 - Resistors
 - Bread board/ Trainer board
 - One multimeter

4. Circuit Diagram:

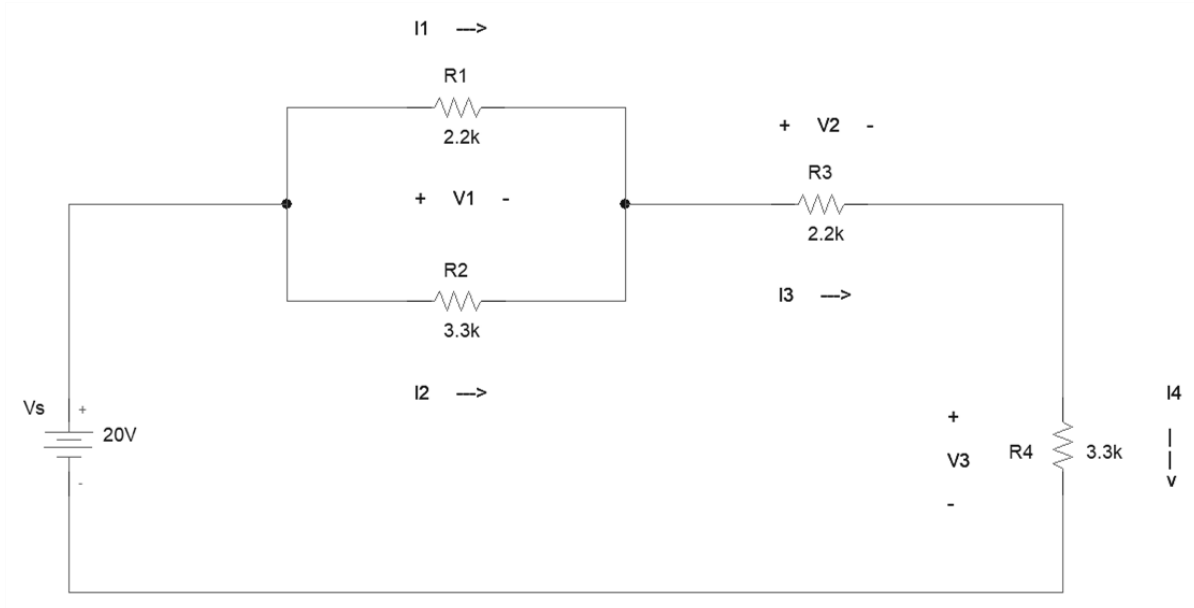


Fig. 1

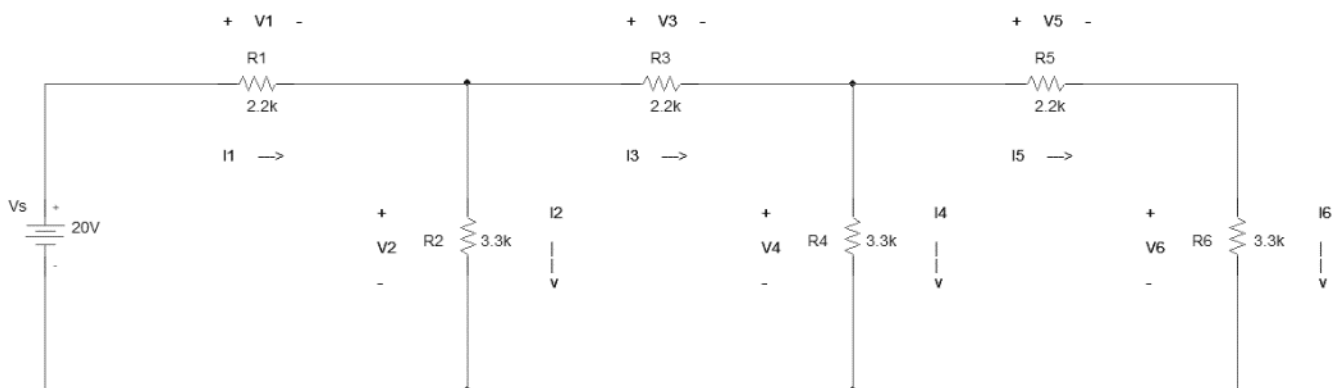


Fig. 2

5. Procedure:

- Set up the circuit as in the above figures.
- Measure the voltage across the resistors and calculate the currents.

6. Data Table:

Fig. 1:

R_1 (k Ω)	R_2 (k Ω)	R_3 (k Ω)	R_4 (k Ω)

V_1 (V)	V_2 (V)	V_3 (V)

I_1 (mA)	I_2 (mA)	I_3 (mA)	I_4 (mA)

Fig. 2:

R_1 (k Ω)	R_2 (k Ω)	R_3 (k Ω)	R_4 (k Ω)	R_5 (k Ω)	R_6 (k Ω)

V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)	V_5 (V)	V_6 (V)

I_1 (mA)	I_2 (mA)	I_3 (mA)	I_4 (mA)	I_5 (mA)	I_6 (mA)

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7. Caution:

1. Don't switch on the supply until the circuit has been checked by your teacher.
2. Take care of the reading of the apparatus.
3. Take care of any bare circuit element in energized condition.

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Experiment No. 3

Verification of Superposition Principle

1. **Objective:** To verify experimentally the Superposition theorem which is an analytical technique of determining currents in a circuit with more than one emf source.
2. **Theoretical Background:** In a linear circuit containing multiple independent sources and linear elements (e.g. resistors, inductors, capacitors) the voltage across (or the current through) any element when all the sources are acting simultaneously may be obtained by adding algebraically all the individual voltages (or the currents) caused by each independent source acting alone, with all other sources deactivated.

An independent voltage source is deactivated (made zero) by shorting it and an independent current source is deactivated (made zero) by open circuiting it. However, if a dependent source is present, it must remain active during the superposition process.

3. Equipment:

- Two DC power supplies adjusted to 10V and 20V
- Resistors: 1k Ω , 2.2k Ω , 3.3k Ω , 4.7k Ω and 5.6k Ω
- Bread board/ Trainer board
- One digital Multimeter (DMM)

4. Circuit Diagram:

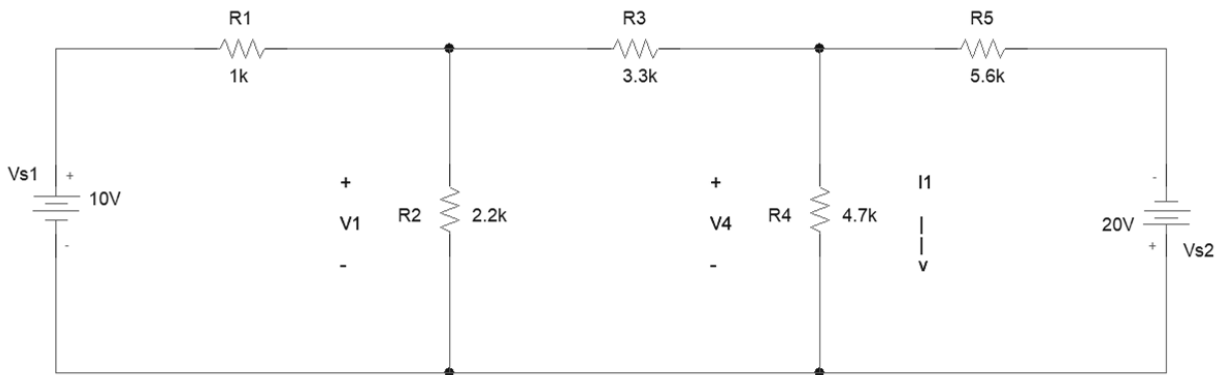


Fig. 1

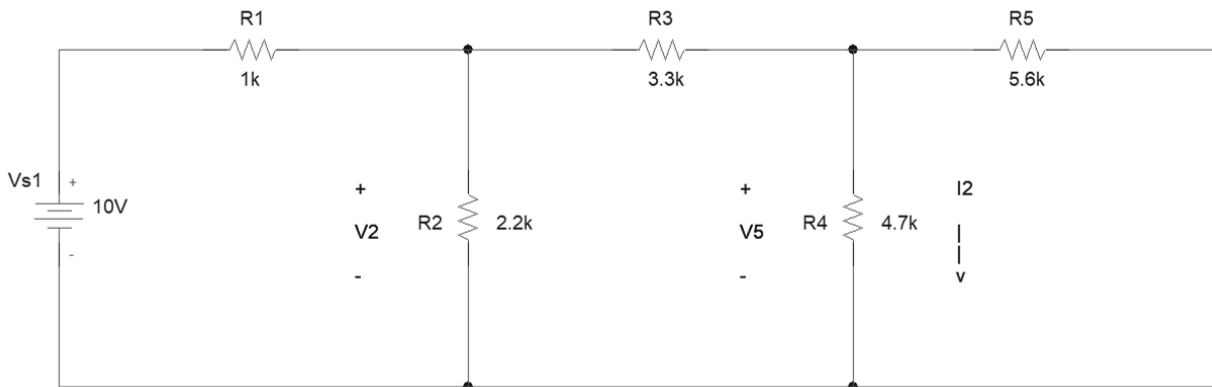


Fig. 2

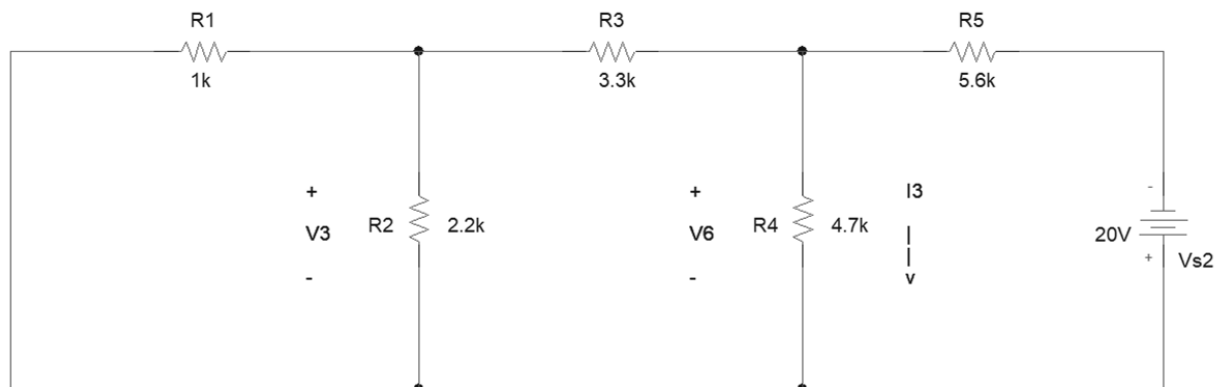


Fig. 3

5. Procedure:

- Setup the circuit as in Fig. 1.
- Measure resistor values accurately with DMM.

- iii. Apply $V_{s1} = 10$ volts from one source and $V_{s2} = 20$ volts from another source. Measure voltages accurately with DMM.
- iv. Measure V_1 with DMM. Measure V_4 and calculate the current I_1 and record it in the given table.
- v. Render V_{s2} inactive (keeping V_{s1} active) and measure V_2 . Measure V_5 and calculate I_2 .
- vi. Render V_{s1} inactive (keeping V_{s2} active) and measure V_3 . Measure V_6 and calculate I_3 .
- vii. Verify if $V_1 = V_2 + V_3$ and $I_1 = I_2 + I_3$ which would validate the superposition theorem for this particular circuit.

6. Data Table:

Fig. 1

R4 (k Ω)	V1 (V)	V4 (V)	$I_1 = V_4 / R_4$ (mA)

Fig. 2

V2 (V)	V5 (V)	$I_2 = V_5 / R_4$ (mA)

Fig. 3

V3 (V)	V6 (V)	$I_3 = V_6 / R_4$ (mA)

Verification

V1 (V)	$V_2 + V_3$ (V)	I_1 (mA)	$I_2 + I_3$ (mA)

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7. Question:

1. "Although Superposition Principle can be used to determine voltage and current in a linear circuit, it cannot be used to determine power." --- Elucidate the statement.

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Experiment No. 5

Verification of Thevenin's Theorem and Maximum Power Transfer Theorem

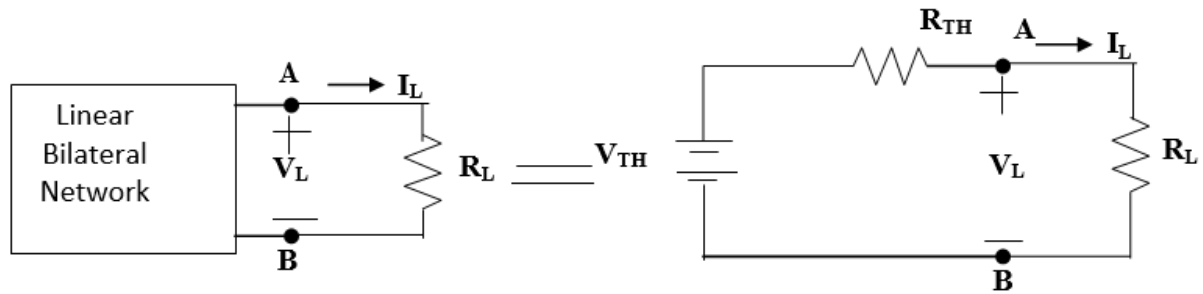
Part-A: Verification of Thevenin's Theorem

1. **Objective:** To verify Thevenin's theorem with reference to a given circuit.
2. **Theoretical Background:** It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all the sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end and it reduces greatly the amount of computations which we have to do each time a change is made. Using Thevenin's theorem, the given circuit, excepting the particular branch to be studied, is reduced to the simplest equivalent circuit possible and then the branch to be changed is connected across the equivalent circuit.

The Thevenin's theorem states that any two terminal linear bilateral network containing sources and passive elements can be replaced by an equivalent circuit consisting of a voltage source V_{th} in series with resistor R_{th} where

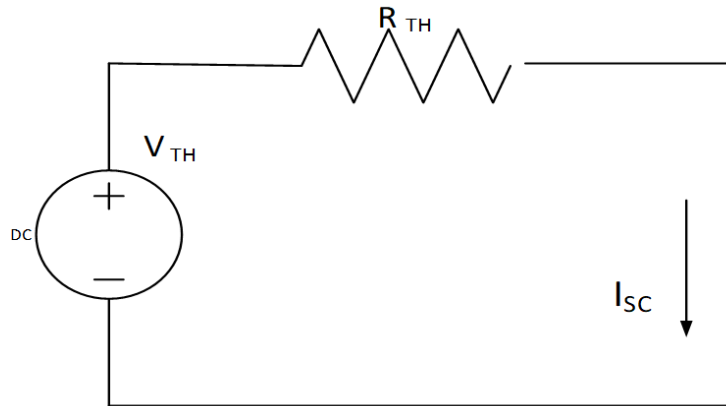
V_{th} = the open circuit voltage (V_{OC}) at the two terminals A & B.

R_{th} = the resistance looking into the terminals A and B of the network with all sources removed.



There are several methods for determining Thevenin resistance R_{TH} . An attractive method for determining R_{TH} is: (1) determine the open circuit voltage, and (2) determine the short circuit current I_{SC} as shown in the figure; then

$$R_{TH} = -\frac{V_{OC}}{I_{SC}}$$



3. Equipment:

- Resistors
- Multimeter
- DC Power Supply adjusted to 20V DC
- Breadboard

4. Circuit Diagram:

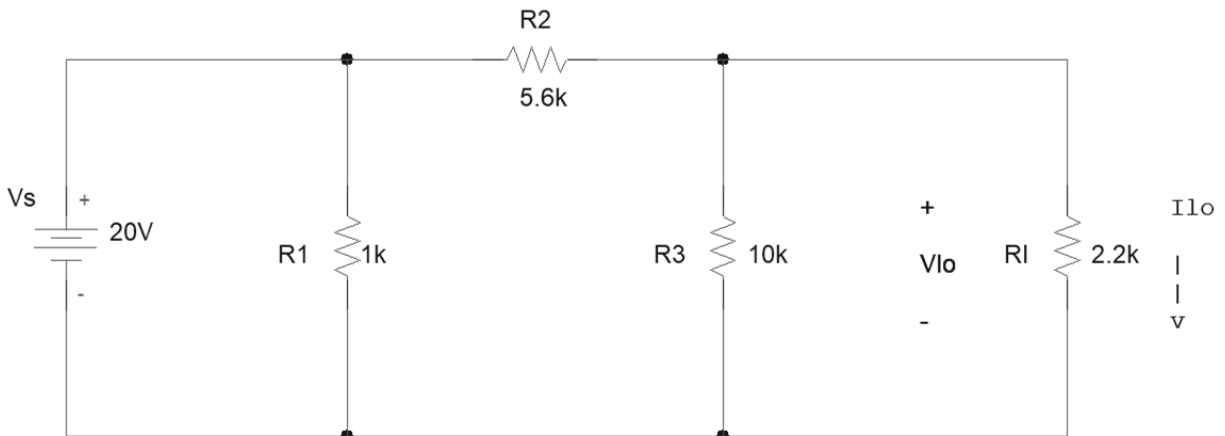


Fig. 1: Original Circuit

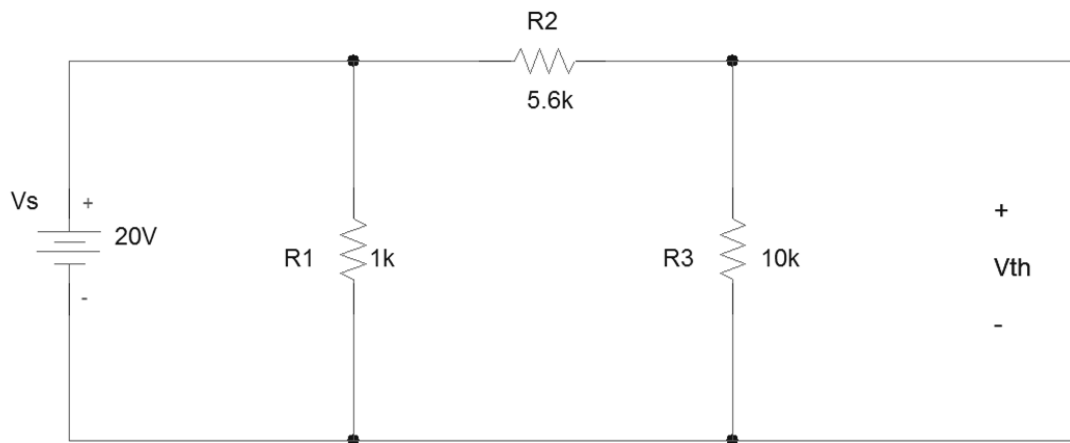


Fig. 2: Circuit for finding V_{th}

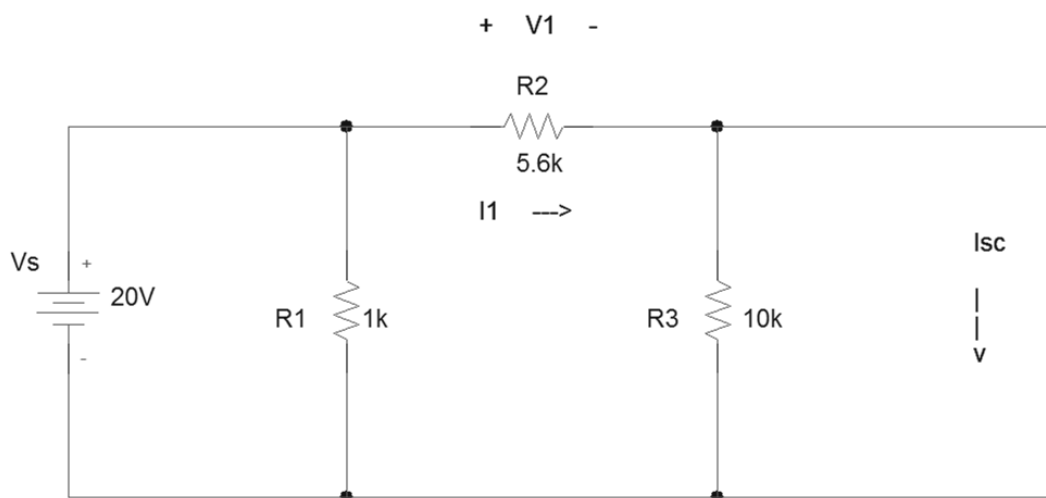


Fig. 3: Circuit for finding I_{sc}

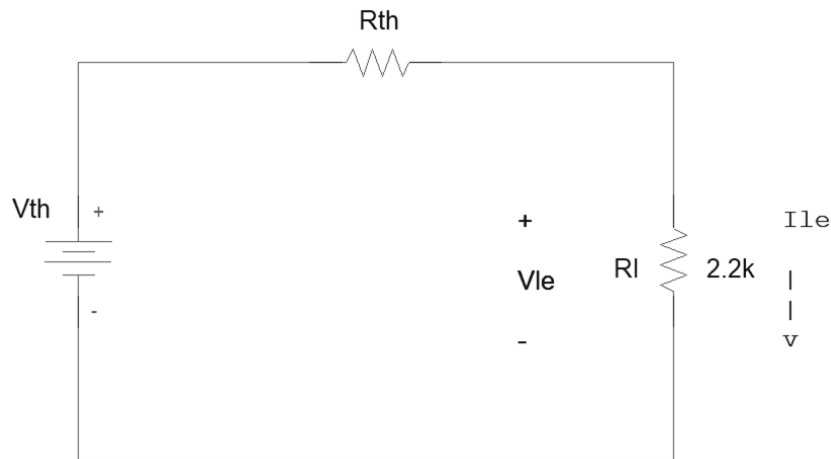


Fig. 4: Equivalent circuit

5. Procedure:

For Original Circuit:

Construct the original circuit as shown in Fig. 1. Connect source voltage $V_s = 20V$ DC from a DC power supply. Measure R_l , V_{l0} , calculate I_{l0} and record the data in the table.

$$I_{l0} = V_{l0}/R_l$$

Finding V_{th} & R_{th} :

- i. Remove the load resistance R_l and find the open circuit voltage. This voltage is Thevenin voltage i.e. V_{th} .
- ii. Place a short circuit between the open terminals and find the short circuit current I_{sc} .

$$R_{th} = V_{th} / I_{sc}$$

$$I_{sc} = I_1 = V_1 / R_2$$

Get back to the circuit in Fig. 2. Deactivate the DC voltage source. Measure the resistance between the open circuit terminals via the multimeter. Verify the earlier value of R_{th} .

For Equivalent Circuit:

Construct the equivalent circuit as shown in Fig. 4 setting the power supply at V_{th} volts and resistor R_{th} as close as possible to the value found from above. Now measure V_{le} and calculate I_{le} . Compare these values with previous values.

6. Data Table:

Table 1: Data for original circuit

R_l ($k\Omega$)	V_{lo} (V)	I_{lo} (mA)

Table 2: Data for V_{th}

V_{th} (V)

Table 3: Data for R_{th}

Indirect Method:

R_2 ($k\Omega$)	V_1 (V)	$I_1 = V_1 / R_2$ (mA)	$I_{sc} = I_1$ (mA)	$R_{th} = V_{th} / I_{sc}$ ($k\Omega$)

Direct Method:

R_{th} ($k\Omega$)

Table 4: Data for equivalent circuit

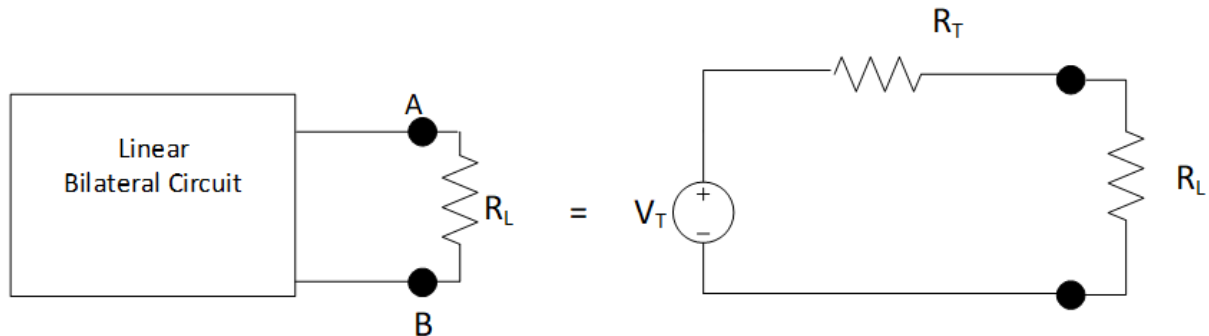
V_{le} (V)	I_{le} (mA)

Faculty Signature and Date

Part-B: Maximum Power Transfer Theorem

1. Objective: The objective of this experiment is to verify maximum power transfer theorem.

2. Theoretical Background: The maximum power transfer theorem states that a resistive load will receive maximum power when its total resistive value is exactly equal to the Thevenin's resistance of the network as seen by the load.



We know that any linear circuit terminated with a load R_L can be reduced to its Thevenin's equivalent. Now according to this theorem, the load R_L will receive maximum power when

$$R_L = R_{th}$$

3. Equipment:

- Digital Multimeter
- DC power supply adjusted to 20V DC
- Resistor and potentiometer
- Wires

4. Circuit Diagram:

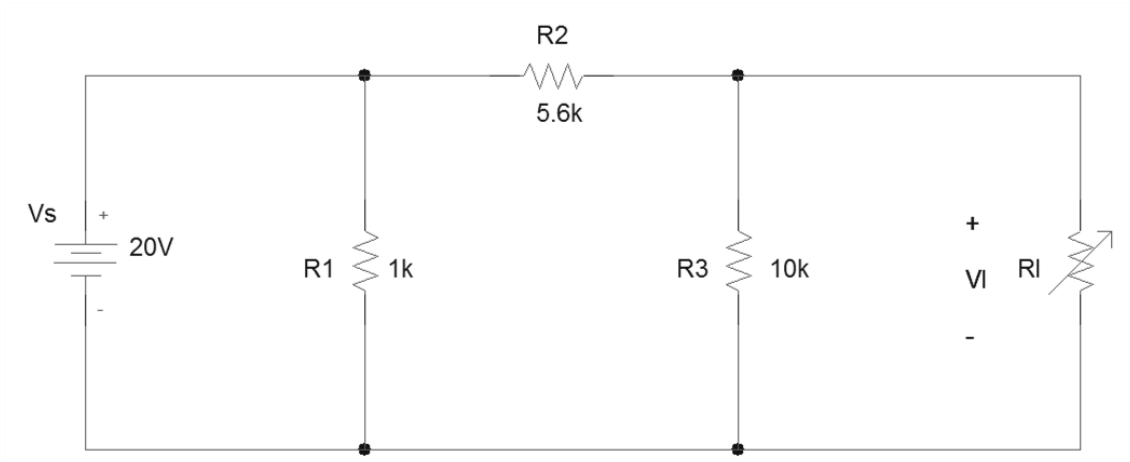


Fig. 5: Circuit for verifying maximum power transfer theorem

5. Procedure:

- Set up the circuit as shown in Fig. 5.
- Use the following values of R_L .

$1k\Omega$, $2.2k\Omega$, $3.3k\Omega$, R_{th} , $4.7k\Omega$, $5.6k\Omega$ and $10k\Omega$

Set R_{th} up using potentiometer. Measure R_L , V_L and calculate $P_L = V_L^2 / R_L$. The power shall be maximum for condition where the $R_L = R_{th}$.

6. Data Table:

R_L ($k\Omega$)	V_L (V)	$P_L = V_L^2 / R_L$ (mW)

Faculty Signature and Date

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Experiment No. 6

Generation and Measurement of Waveforms

1. **Objective:** The objective of this experiment is to get introduced to the ac voltage source and the oscilloscope.
2. **Theoretical Background:**

AC Voltage Source

It generates an ac voltage.

It can deliver 3 waveforms. They are square, triangular and sinusoidal.

The frequency of the voltage can be set up by selecting a frequency range and turning the frequency knob.

The amplitude of the voltage can be set up by turning the amplitude knob.

A dc voltage can be added to the ac voltage by pulling out the offset knob and turning it.

Oscilloscope

It shows the graph of a voltage. The x axis shows time and the y axis shows voltage.

The screen – It looks like a graph paper. The horizontal line at the middle is the x axis. The vertical line at the middle is the y axis.

The CAL terminal – It may be used to check if the probes are functional or not.

The FOCUS knob – The focus should be such that the trace is thin and the readings are sharp.

The INTEN knob – The brightness of both the trace and readings should be low.

The channels – The oscilloscope has 2 channels. They are channel 1 and 2. Each channel can show a graph. The screen can show 1 channel at a time or both simultaneously.

The probes – The oscilloscope has 2 probes. Each goes with a channel. Each has 2 terminals. One terminal looks like pen and another terminal is a clip. The pen terminal is for + terminal of the voltage and the clip terminal is for - terminal of the voltage. The clip terminals of both probes are joined together internally. Each pen terminal has a switch with 2 options. They are $\times 1$ and $\times 10$. The $\times 1$ option should be selected for the general purpose.

The GND button – The voltage reading is 0 V when the probes are not connected to the circuit. It is shown as a horizontal line. Each channel has a button to bring up the zero-line.

The AC/DC button – Each channel has two types of input coupling. They are ac and dc. The ac coupling shows only the ac voltage component of the input. The dc coupling shows everything.

The CH1 and CH2 button – Each channel can be turned on or off.

The VOLTS/DIV knob – It selects the vertical scale of the screen for each channel. The channels can have different scales. If both are shown simultaneously on the screen, they should have the same scale.

The ALT/CHOP/ADD and INV button – The ALT/CHOP/ADD button works when both channels are shown simultaneously on the screen. The ALT option should be selected for the general purpose. If the graphs flickers much, the CHOP option should be selected. The INV button can be long pressed to invert channel 2. Channel 1 cannot be inverted. Inversion shifts the zero-line automatically.

The VERTICAL POSITION knob – It shifts the zero-line up or down for each channel. It is used for the purpose of viewing only.

The TIME/DIV knob – It selects the horizontal scale of the screen for both channels.

The MAIN/ALT/DELAY button – The MAIN option should be selected for the general purpose.

The HORIZONTAL POSITION knob – It shifts the zero-line left or right for both channels. It is used for the purpose of viewing only.

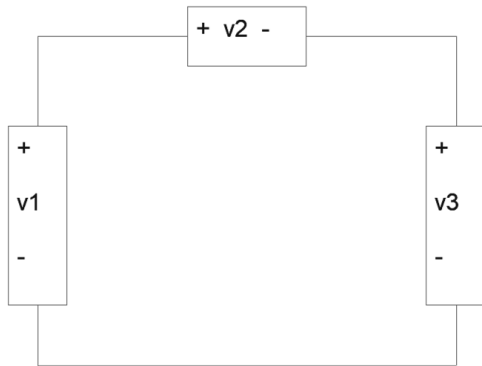
The TRIGGER LEVEL knob – The value should be 0 V for the general purpose.

The MODE button – The ATO option should be selected for the general purpose.

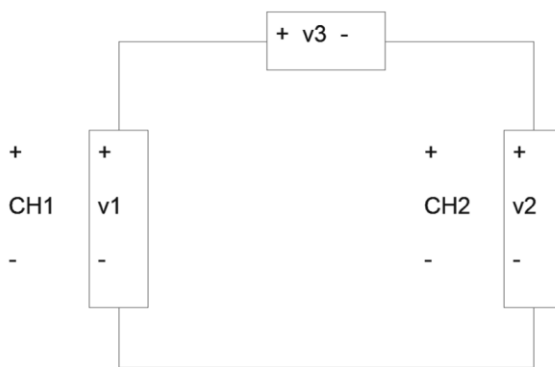
The SOURCE button – The CH1 option should be selected when channel 1 or both channels are shown on the screen. The CH2 option should be selected when channel 2 is shown on the screen.

The COUPLING button – The AC option should be selected for the general purpose.

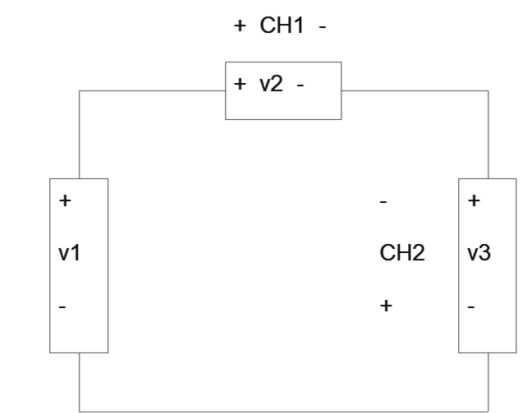
Showing two graphs simultaneously – Let us consider the following circuit.



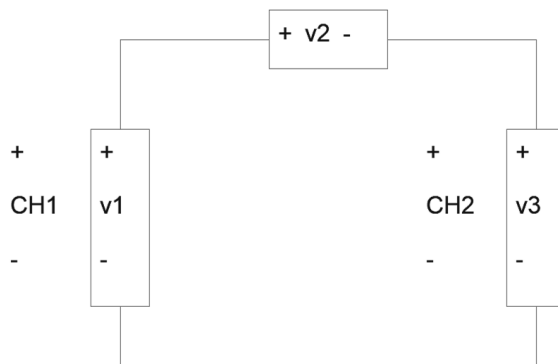
If v_1 and v_2 are to be shown, then the channels should be connected the following way.



If v_2 and v_3 are to be shown, then the channels should be connected the following way. Channel 2 should be inverted.



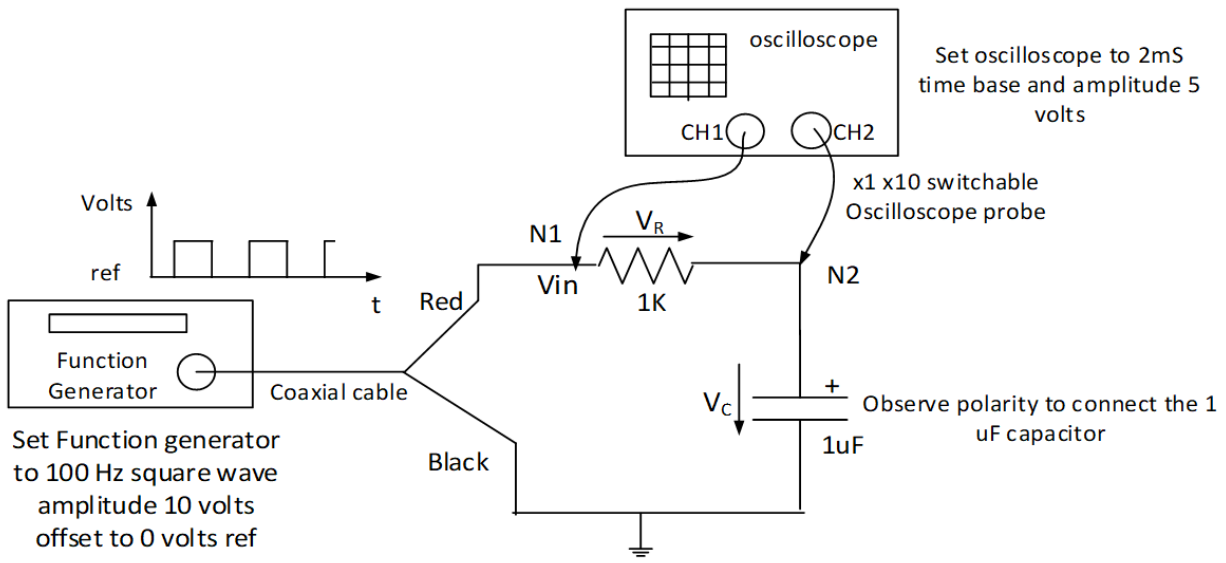
If v_1 and v_3 are to be shown, then the channels should be connected the following way.



3. Equipment:

- Resistance: $1\text{k}\Omega$
- Capacitance: $1\mu\text{F}$
- Oscilloscope and probes
- Signal Generator (Function Generator) and cables
- Wires
- Breadboard

4. Experimental Setup:



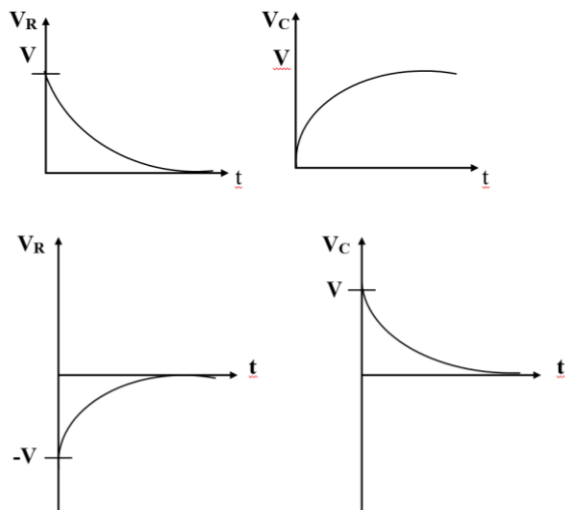
Experimental setup to observe transient behaviour of a RC circuit

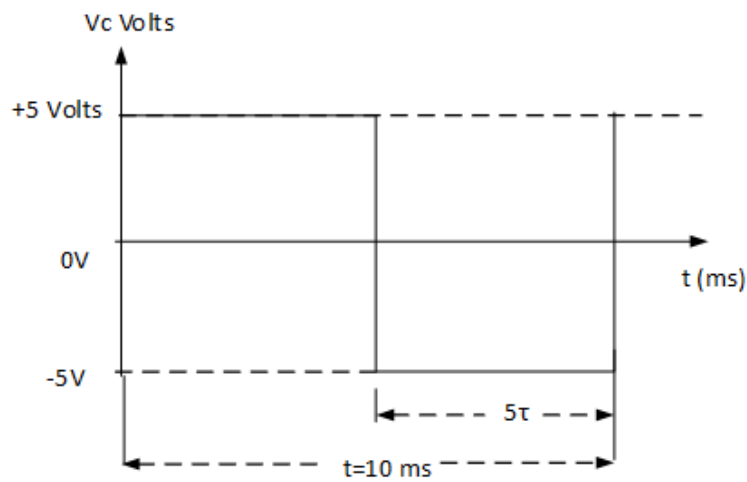
Fig. 1

5. Procedure:

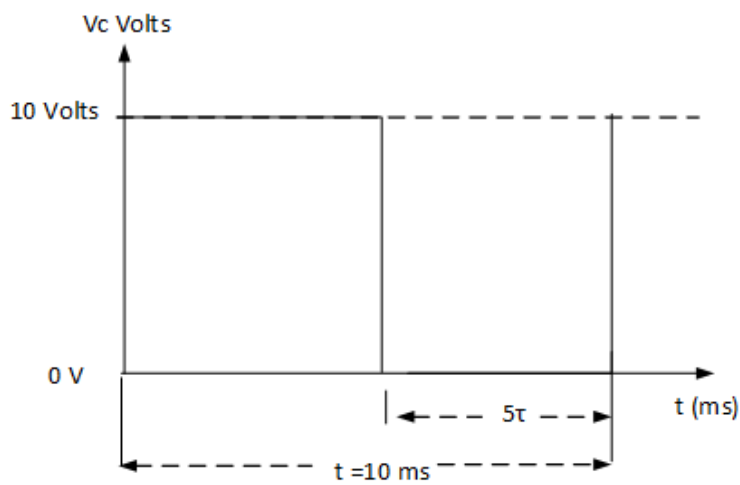
- Setup the circuit as shown in Fig. 1.
- Apply 100Hz square wave from signal generator/function generator.
- Observe all wave shapes at Ch.1.
- Observe all wave shapes at Ch.1 and Ch.2.

6. Waveforms:

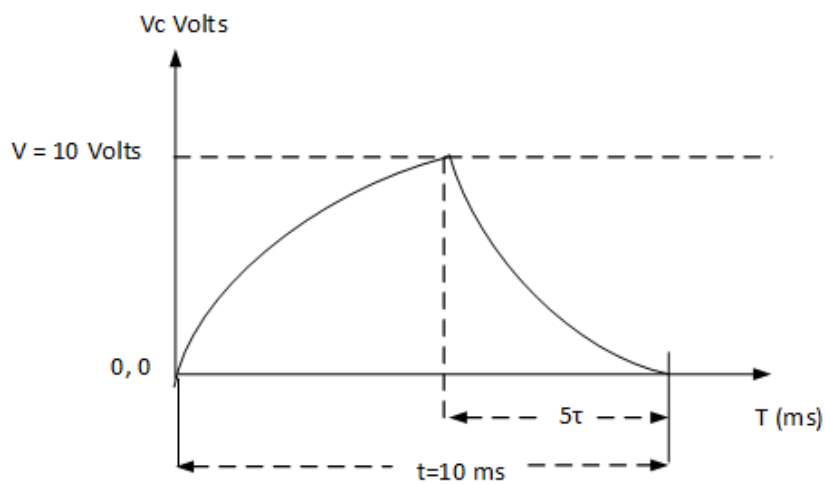




An NRZ Square wave with frequency of 100 Hz with amplitude +5 Volts
- 5 Volts volts along the reference axis
 $\tau = 1\text{ms}$



Square wave = 100 Hz with amplitude 10 volts offset adjusted above the reference axis



Waveform showing charging and discharging of an RC circuit

$$\tau = RC = 1 \times 10^3 \times 1 \times 10^{-6} = 1 \text{ ms}$$

Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic Engineering (EEE)

Brac University

Experiment No. 7

Study of Transient Behavior of RC Circuit

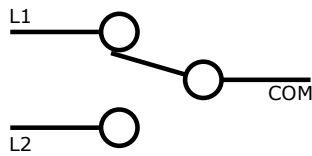
1. **Objective:** The objective of this experiment is to study Transient Response of RC circuit with step Input. In this experiment we shall apply a square wave input to an RC circuit and observe the respective wave-shapes and determine the time constants.
2. **Theoretical Background:** A switch is an electrical component that can disconnect or connect the conducting path in an electrical circuit, interrupting the electric current or diverting it from one conductor to another. The most common type of switch is an electromechanical device consisting of one or more sets of movable electrical contacts connected to external circuits. When a pair of contacts is touching each other, current can pass between them. When the contacts are separated, no current can flow.

The most familiar form of switch is a manually operated electromechanical device with one or more sets of electrical contacts, which are connected to external circuits. Each set of contacts can be in one of two states: either "closed" meaning the contacts are touching and electricity can flow between them, or "open", meaning the contacts are separated and the switch is nonconducting.

SPST Switch: A single-pole, single-throw switch is a simple on-off switch. The two terminals are either connected together or disconnected from each other. An example is a light switch.

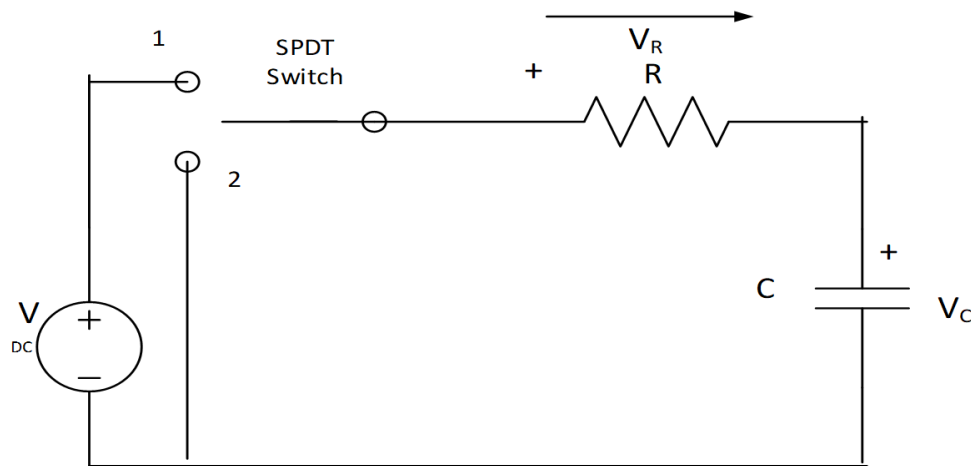


SPDT Switch: A single-pole, double-throw switch gets its C terminal (COM, Common) connected either to L1 terminal or L2.



The transient response is the temporary response that results from a switching operation and disappears with time. The steady state response is that which exists after a long time following any switching operation.

Let us consider an RC circuit shown in figure.



Charging Phase:

When the switch is connected to position 1, applying KVL we can write

$$V = Ri + \frac{1}{C} \int i dt \dots \dots \dots (1)$$

If the capacitor is initially uncharged, the solution of equation (1) is

$$i = \frac{V}{R} e^{-\frac{t}{\tau}} \dots \dots (2)$$

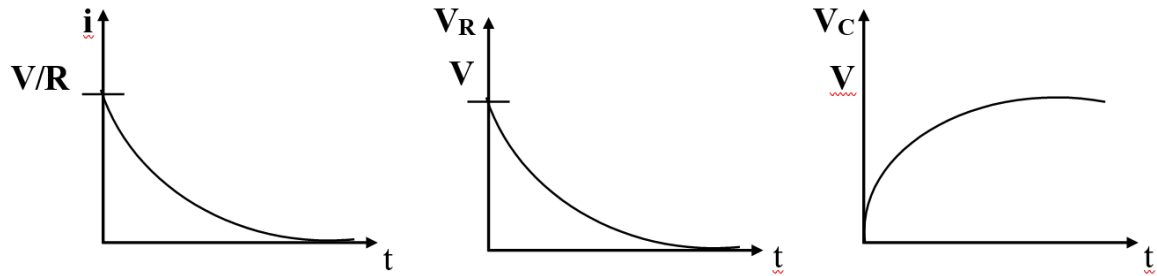
Therefore, the voltage across the resistor and capacitor are given by

$$V_R = V e^{-\frac{t}{\tau}} \dots \dots \dots (3)$$

$$V_C = V - V_R = V (1 - e^{-\frac{t}{\tau}}) \dots\dots (4)$$

Where $\tau = RC$ and is called the time constant of the circuit.

Equations (2), (3) & (4) are plotted below:



It is seen from the curves that the voltage across the capacitor rises from zero to V volts exponentially and the charging current is maximum at the start i.e. when C is uncharged, then it decreases exponentially and finally ceases to zero when the capacitor voltage becomes V .

Discharging Phase:

When the switch is connected to position 2, applying KVL we can write

$$0 = Ri + \frac{1}{C} \int i dt \dots\dots\dots (5)$$

Since the voltage across the capacitor is now V , the solution of equation (5) is

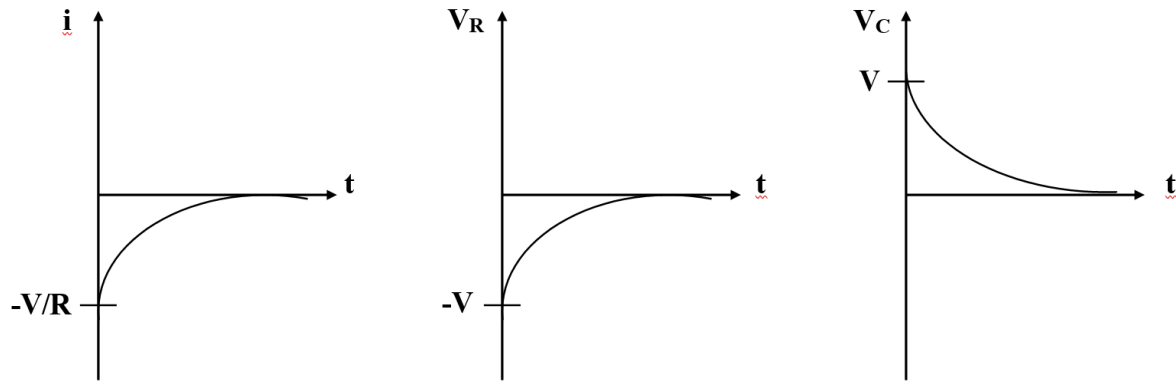
$$i = -\frac{V}{R} e^{-\frac{t}{\tau}} \dots\dots\dots (6)$$

Therefore, the voltage across the resistor and capacitor are given by

$$V_R = -V e^{-\frac{t}{\tau}} \dots\dots\dots (7)$$

$$V_C = V e^{-\frac{t}{\tau}} \dots\dots\dots (8)$$

Equations (6), (7) & (8) are plotted below:



It is seen from the curves that the voltage across the capacitor falls from V to zero volts exponentially. The charging current is maximum at the start i.e. when the switch is just thrown to position 2, then it decreases exponentially and finally ceases to zero when the capacitor voltage becomes zero.

Signal Generator / Function Generator:

Select the Square wave button, select the 100 Hz range and adjust the display to show 100 Hz. Adjust the offset knob (pull to enable) so that the square wave reference level is on the reference line.

Oscilloscope:

Select Channel 1 to display a single straight line in the screen. Select the x1 in the probe. Adjust Volts per division knob to show 5 Volts/div and time $t = 2$ ms.

Select trigger to – Auto, coupling to ac and coupling to Channel 1.

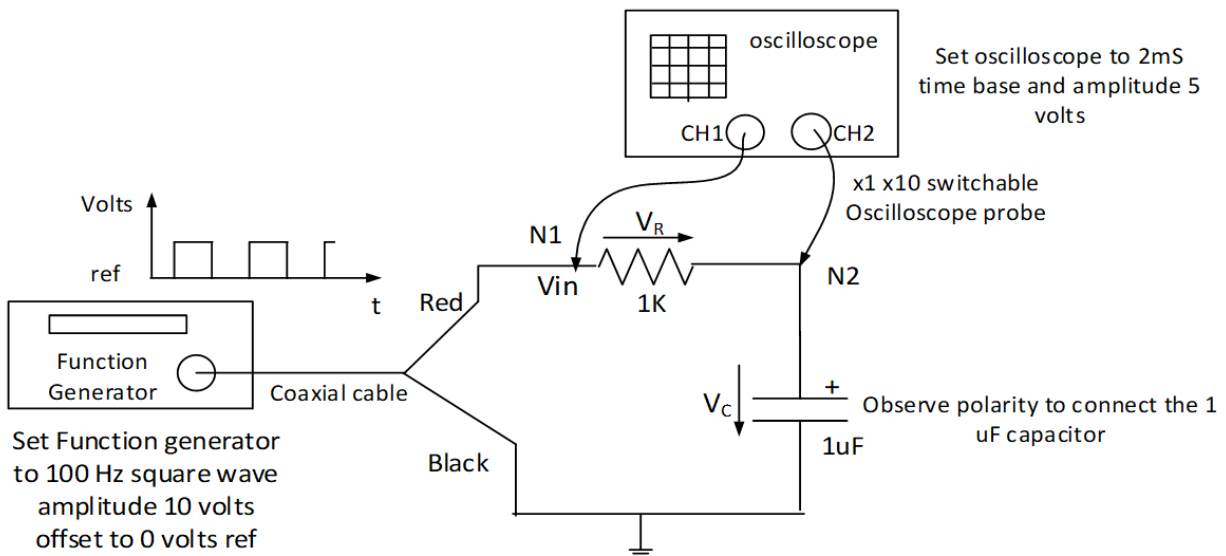
Connect signal generator to the oscilloscope probe and adjust the amplitude control of the signal generator to adjust the amplitude of the square wave to 10 volts. The oscilloscope should now display in Channel 1 an NRZ square wave with an amplitude of 10 volts and a time period of 10 ms along the reference axis. Now pull out the offset adjustment knob and adjust the waveform so that it is along the reference axis.

Switch on Channel 2 so now you can observe both traces. Adjust Channel 2 in the oscilloscope the same way to $t = 2$ ms and Voltage to read 5 V/div. Connect Channel 1 and Channel 2 as shown above.

3. Equipment:

- Resistance: $1\text{k}\Omega$
- Capacitance: $1\mu\text{F}$
- Oscilloscope and probes
- Signal Generator (Function Generator) and cables
- Wires
- Breadboard

4. Experimental Setup:



Experimental setup to observe transient behaviour of a RC circuit

Fig. 1

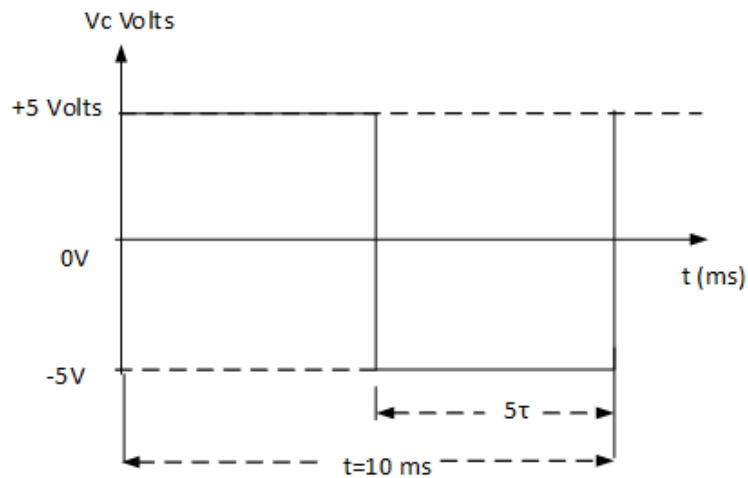
5. Procedure:

- Setup the circuit as shown in Fig. 1.
- Apply 100Hz square wave from signal generator/function generator.
- Observe all wave shapes at Ch.1 and draw them.
- Observe all wave shapes at Ch.1 and Ch.2 and draw them.

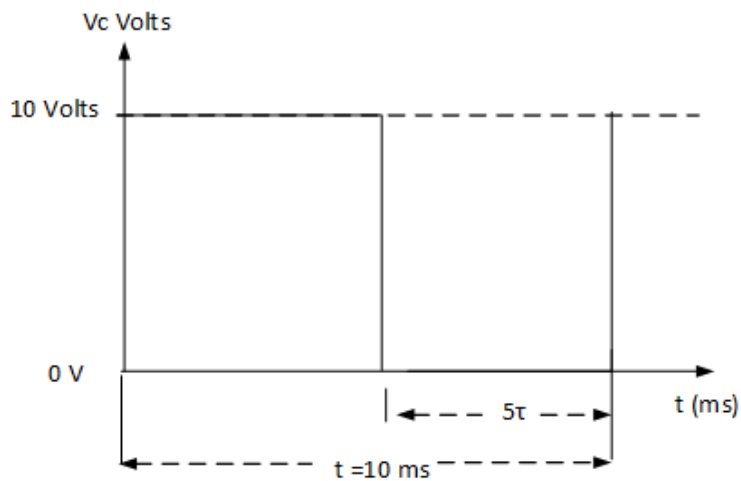
6. Question:

Define time constant for an RC circuit. What is the significance of time constant? How time constant can be determined?

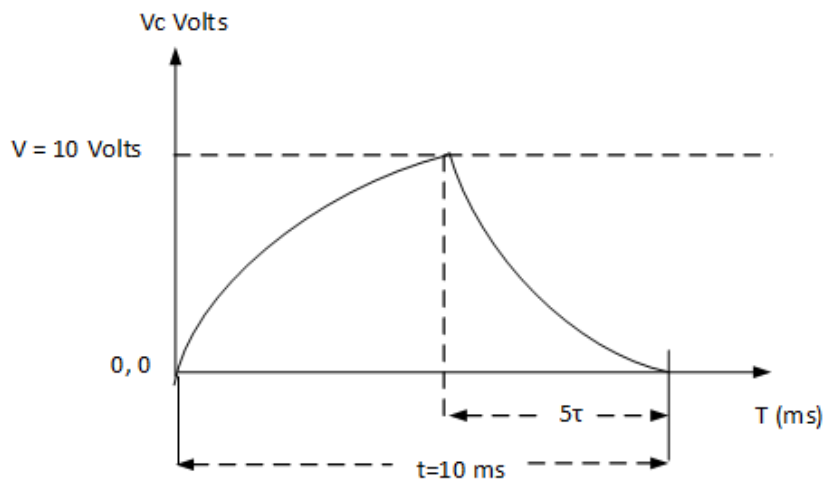
7. Waveforms:



An NRZ Square wave with frequency of 100 Hz with amplitude +5 Volts - 5 Volts along the reference axis
 $\tau = 1 \text{ ms}$



Square wave = 100 Hz with amplitude 10 volts offset adjusted above the reference axis



Waveform showing charging and discharging of an RC circuit

$$\tau = RC = 1 \times 10^3 \times 1 \times 10^{-6} = 1 \text{ ms}$$

Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic Engineering (EEE)

Brac University

Project

Total Marks: 25

Time: 2 Weeks

1. **Objective:** Design a circuit which will give a fluctuating DC Output voltage for an Input Square Wave.
2. **Problem Statement:** You will have a square wave as your source voltage where the square wave will have a Peak Amplitude of $(19+G)$ V and Minimum Amplitude as 0 V with a frequency of $(20 \times G)$ Hz. Where G is your Group Number. The output voltage is across a capacitor which will fluctuate. Besides, the driving voltage of the capacitor should be another square wave having a Peak Amplitude of 10 V and Minimum Amplitude as 0 V with the same frequency. The time constant is 1ms. Now, design the whole circuit. Show the input/output graph in PSpice for the first 10 cycles.
3. **Submission Guidelines:**
One group should submit one simulation (.sch) file and one report (.pdf) file. The report must contain the following:
 1. Cover Page
 2. Objective (write the problem statement here)
 3. Design (detailed calculation of values for all the selected components) (explain how you are getting the driving voltage of the capacitor from the source voltage. Hint: Show that Thevenin equivalent circuit from the source voltage gives the driving voltage of the capacitor where the capacitor is acting as load.)
 4. PSpice Circuit Diagram
 5. Input / Output Waveforms

4. Marks Distribution:

Design	10
PSpice Circuit Diagram	10
Input / Output Waveforms	5
Total	25